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RESEARCH

Validated HPLC-FLD Method with Pre-column Derivatization for the Simultaneous Determination of Glutathione and Tranexamic Acid and Its Application to Ex Vivo Skin Permeability from w/o and o/w Nano-emulsions

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Abstract

Purpose Melasma is characterized by dark spots on the skin, particularly on the face, caused by hormonal changes, sun exposure, and genetic predisposition. Glutathione (GSH) and tranexamic acid (TXA) can help prevent this discoloration. This study aims to develop and validate an HPLC-FLD method for simultaneous quantification of both analytes in complex matrices and to investigate the influence of the nano-emulsion type in their skin permeation.

Methods A C18 column (15 cm × 4.6 mm, 5 μm) was used as the stationary phase, while a mixture of methanol and phosphate buffer (pH 6.7) served as the mobile phase in a gradient elution system. For fluorescence detection, pre-column derivatization was performed using ortho-phthalaldehyde and N-acetyl-L-cysteine in borate buffer (pH 10.5). Reaction conditions and product stability were also evaluated. Two emulsions (o/w and w/o) were formulated and evaluated in terms of oil phase, droplet size and zeta potential. A Salt-Assisted Liquid-Liquid Extraction (SALLE) approach was optimized for nano-emulsions pretreatments. An ex vivo permeability study was performed using Franz diffusion cells and porcine skin.

Results The method showed excellent linearity ($R^2 > 0.999$), sensitivity ($LOD_{TXA} = 3.9$ ng/mL, $LOD_{GSH} = 2.6$ ng/mL), and reliability ($\%RSD_{TXA} < 2\%$, $\%RSD_{GSH} < 2.5\%$, %Recovery = 100 ± 4%). The SALLE procedure provided clean extracts and high %Recoveries. Tape stripping studies and permeation results showed that w/o emulsions significantly enhance TXA uptake and penetration, whereas GSH showed limited permeation regardless of formulation, consistent with both its physicochemical properties and its chemical instability under the experimental conditions.

Conclusion These findings highlight the importance of selecting vehicle specific to each molecule to achieve optimal topical performance.

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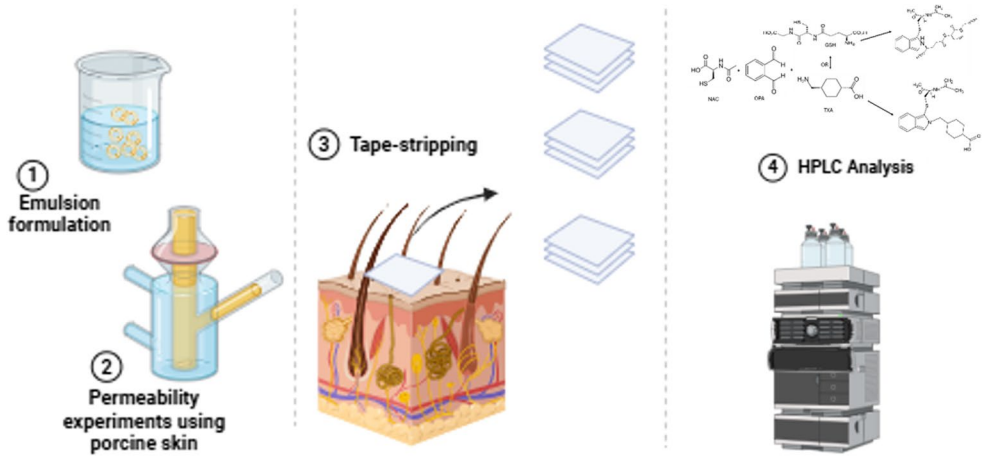
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Graphical Abstract



Keywords Melasma · Glutathione · Tranexamic acid · HPLC- pre-column derivatization FLD · *ex-vivo* permeability study · Tape stripping

Introduction

Melasma is a common, chronic, and relapsing skin disorder that affects sun-exposed areas and is characterized by brown or grayish-brown patches, typically appearing on the face (cheeks, nose, chin, and upper lip) [1]. It results from the activity of melanocytes, which produce and store the dark pigment melanin. In response to light, heat, ultraviolet radiation, or hormonal stimulation, melanocytes may overproduce melanin, leading to the formation of hyperpigmented lesions [2].

Between 1.5% and 33% of the global population may develop melasma, with the condition occurring most frequently in women of reproductive age (20–40 years) and particularly in pregnant women (15–50%). As the pathogenesis of melasma is not yet fully understood, no definitive treatment exists that can completely eliminate the disorder. Nevertheless, new therapeutic strategies may be developed to target pathways involved in persistent pigmentation and in the removal of melanin from the epidermis. Tranexamic acid (TXA), an antifibrinolytic agent, reduces melanin production by modulating the activity of enzymes involved in melanogenesis [3].

L-Glutathione, also known as reduced glutathione or simply GSH, is a tripeptide composed of three amino acids: L-glutamic acid, L-cysteine, and glycine. It is present in nearly all cells and, in its reduced form (GSH), functions as a potent antioxidant. GSH plays a key role in protecting cells from oxidative stress by neutralizing reactive oxygen species such as free radicals and peroxides. The thiol (-SH) group of its cysteine residue acts as a reducing agent and can undergo reversible oxidation and reduction [4].

The combination of reduced glutathione and tranexamic acid has been shown to enhance treatment effectiveness, contributing to a greater reduction in the extent of chloasma [5]. On the global market, TXA is available either alone or in combination with GSH in various formulations intended for systemic (oral, intradermal injection) or topical administration (eye creams, serums, ointments). Typical tablet dosages range from 50 to 250 mg for GSH and up to 250 mg for TXA (e.g., Anthocyn®), while topical creams usually contain both APIs at concentrations of approximately 1,000 ppm (e.g., Mary & May Eye Cream®). Both APIs are commonly administered epidermally, as topical application delivers them directly to the target receptors while bypassing systemic circulation.

The suitability of epidermal formulations is generally assessed through in vitro and ex vivo permeability studies. These procedures require appropriate equipment, such as Franz diffusion cells, along with artificial membranes or human/animal skin tissues [6, 7].

The increasingly widespread use of such preparations makes their reliable large-scale production essential in order to ensure consistent qualitative and quantitative characteristics. Therefore, the development of a validated, flexible, and robust method for the simultaneous quantitative determination of the two active components, TXA and GSH, is necessary. Structurally, both substances lack π-electrons in their molecules and therefore cannot act as chromophores or fluorophores, resulting in poor signal response in ultraviolet or fluorescence spectroscopy [8]. Therefore, their quantification using universal detectors such as Evaporative Light Scattering (ELSD) or Refractive Index (RI) could be applied, but only for routine analyses due to their limited sensitivity

(mg/mL range). More sensitive and effective detectors, such as mass spectrometry (LC-MS/MS), are available; however, because both analytes have low molecular weight, their limits of quantification (LOQ) are not particularly low, while the overall cost of these methods remains high [9].

In several studies, the determination of TXA has been achieved through derivatization with suitable reagents, such as phenylisothiocyanate, ninhydrin [10], sodium picrylsulfonate [11], 2-hydroxynaphthaldehyde [12], or benzenesulfonyl chloride [13], followed by HPLC analysis with UV detection. For fluorometric detection combined with HPLC, derivatization with naphthalene-2,3-dicarboxaldehyde and cyanide, or with o-phthalaldehyde (OPA) and mercaptoethanol, has also been reported [14, 15]. Additionally, TXA derivatives have been quantified using LC-MS [16], UPLC-MS/MS [17], and GC [18] methods.

For GSH determination, the United States Pharmacopeia (USP) monograph specifies an iodine titration method [19]. However, because iodine titration cannot distinguish GSH from its three impurities (A, B, and D), it may yield assay values higher than the true concentration. The European Pharmacopoeia (EP) monograph employs a capillary electrophoresis (CE) method for the detection and quantification of GSH and its related impurities A-D and impurity E [20]. Similar HPLC and CE-based techniques have also been reported by other research groups [21–23]. Alternatively, a qNMR method has been reported for the qualitative and quantitative analysis of GSH and its impurities [24]. However, even in this case, a derivatization reaction is required to enhance detectability. A commonly used reagent for thiol derivatization is mono-bromobimane [25].

Although several methods exist for the quantitative analysis of TXA and GSH individually, no bibliographic reference has been identified for their simultaneous analysis in dosage forms or complex samples. The innovation of the present study lies in their concurrent determination using HPLC-FLD with pre-column derivatization by OPA/NAC. For the purposes of the study, two epidermal nano-emulsion systems (o/w and w/o) incorporating both active substances were prepared and used as carriers for the simultaneous topical delivery of GSH and TXA. Both systems were characterized in terms of the composition of the oily phase, droplet size, and zeta potential, while their suitability as carriers was further evaluated through ex vivo transdermal permeability studies (P_{app}) using Franz diffusion cells and porcine skin, combined with tape-stripping experiments.

To assess the stability of the active substances under conditions relevant to topical application, GSH and TXA were incubated in PBS and in both nano-emulsions at 37 °C for 24 h, revealing temperature-dependent degradation more pronounced for GSH. A one-month stability study at 4 °C further confirmed that both nano-emulsions maintained stable droplet size, zeta potential, and active-substance content, supporting their suitability as topical delivery carriers.

Overall, this combined analytical and pharmaceutic approach provides a sensitive and cost-effective tool for the simultaneous quantification of TXA and GSH and supports the rational development of effective nano-emulsions for the management of hyperpigmentation.

Materials and Methods

Chemicals and Solutions

Acetonitrile (ACN) and methanol, both HPLC gradient grade, were obtained from VWR Chemicals (Radnor, USA). High-purity water (18.2 MΩ·cm resistivity) was produced using a B30 water purification system (Adrona SIA, Riga, Latvia). NaH_2PO_4 and 1 N NaOH (Merck, Darmstadt, Germany) were used for the preparation of the buffer solution in the mobile phase.

For the derivatization reaction, 0.02 M stock solutions of o-phthalaldehyde (OPA) (Sigma-Aldrich, Steinheim, Germany), N-acetyl-L-cysteine (NAC) (Sigma-Aldrich, St. Louis, USA), 3-mercaptopropionic acid (3-MPA) (Sigma-Aldrich, St. Louis, USA), 2-mercaptoethanol (2-ME) (Ega-Chemie, Steinheim, Germany), and tiopronin (Thiola) (Sigma-Aldrich, Steinheim, Germany) were prepared. A borate buffer (pH 10; 700 mM) (Merck, Darmstadt, Germany) was used as the incubation medium to prevent ionization of secondary amino groups. All solutions were prepared in water, except for the OPA reagent, which was dissolved in methanol.

Calibration curves were constructed using freshly prepared aqueous mixed-standard solutions containing 0.5 mg/mL TXA (Sigma-Aldrich, Steinheim, Germany) and 0.5 mg/mL GSH (Sigma-Aldrich, Steinheim, Germany). Appropriate dilutions were then made to obtain six concentration levels ranging from 7.5 to 375 ng/mL (after derivatization). Following derivatization, all samples were analyzed by HPLC.

The two emulsions (w/o and o/w) were prepared using Fractionated Coconut Oil with Caprylic/Capric/Lauric triglycerides, *Prunus armenica* kernel oil, and *Prunus amygdalus* dulcis oil (AROMA LAB, Thessaloniki, Greece) as the oil phase. A mixture of Tween 80 (Chemco, Athens, Greece) and Span 80 (Sigma, St. Louis, USA) was used as the surfactant system.

For the ex vivo permeability studies, a phosphate-buffered saline (PBS) solution (pH 7.4) was prepared by dissolving 1.44 g NaH_2PO_4 (Merck, Darmstadt, Germany), 0.24 g KH_2PO_4, 8.0 g NaCl (Sigma-Aldrich, Steinheim, Germany), and 0.20 g KCl (Chem-Lab NV, Zedelgem, Belgium) in 1 L of distilled water. Porcine skin was obtained fresh from a local slaughterhouse immediately following

the animal's death. Samples were harvested from the flank region (regio abdominis media/fossa paralumbalis) and gently rinsed with PBS to remove residual blood. Hair was removed using scissors prior to further processing. The skin was dermatomed to a uniform thickness of approximately 0.7 mm using a Zimmer® electric dermatome (Zimmer Surgical, Inc., Dover, Ohio, USA; blade REF 00-8800-000-10). Skin samples were used fresh on the day of collection.

Instrumentation

Chromatographic separation was performed using a Shimadzu HPLC system consisting of a DGU-14 A degasser, two LC-20AD pumps, a CTO-10AS VP column oven (Shimadzu, Tokyo, Japan) set at 30 °C, a SIL-10AD autosampler (cooled rack at 4 °C) with injection volume: 30 µL), and an RF-20 A fluorescence detector (Shimadzu, Tokyo, Japan). The detector was operated at Gain ×4 and medium sensitivity. Both analytes were monitored at $\lambda_{exc}/\lambda_{em}$ = 325/425 nm. A Supelco HS Discovery C18 reversed-phase column (15 cm × 4.6 mm, 5 µm) was used as the stationary phase.

Ultrasonic Processors VSX 130 (Athens, Greece) and an Ultra-Turrax homogenizer (IKA, Staufen im Breisgau, Germany) were employed for the preparation of the emulsions. A Zetasizer (Malvern ZEN3600, Malvern, Worcestershire, United Kingdom) was used to determine zeta potential and droplet size. The rheological properties of the formulations were evaluated using a Rheometer (Anton Paar MCR 92, Graz, Austria).

Franz diffusion cells were used to assess the transdermal permeability of TXA and GSH incorporated into the o/w and w/o emulsions. Porcine epidermis was mounted between the donor and receptor compartments (5 mL), providing an effective diffusion area of 0.2 cm². Phosphate buffered saline (PBS, pH 7.4) was used as the receptor medium. The temperature was maintained at 37 °C by circulating warm water through the double-jacketed chamber, and gentle agitation was applied at 70 rpm.

Nano-emulsion Preparation Process

For the execution and completion of the experimental procedures, two types of emulsions (w/o and o/w) were prepared. To ensure comparable results, the same excipients were used in both formulations, with their ratios adjusted according to the emulsion type, in accordance with the relevant literature [26]. Specifically, the o/w emulsion was prepared using a ratio of 80:10:10 v/v (water: oil: surfactant), whereas the w/o emulsion followed a 40:50:10 v/v composition.

To achieve optimal characteristics, such as small droplet size and appropriate zeta potential, both critical for emulsion stability, the Ultrasound Vibra-Cell Processor was employed. Initially, under continuous stirring at 25 °C, the oil phase was gradually added to the aqueous. As the surfactant system, a mixture of Tween 80 and Span 80 at a 5:95 v/v ratio was used, with Span 80 incorporated into the oil and Tween 80 into the aqueous phase.

Derivatization Procedure

According to the derivatization procedure, 450 µL of the sample solution (TXA and GSH) were mixed with 50 µL of OPA (0.02 M), 50 µL of NAC (0.02 M), and 50 µL of boric acid buffer (pH 10.5; 700 mM) in a glass HPLC vial. The mixture was vortexed for 15 s, allowed to react for 120 min at room temperature, and subsequently injected into the HPLC-FLD (cooled rack at 4 °C) system. After completion of the reaction, sample stability could be maintained for at least 15 h at 4 °C by adding 125 µL of NaOH (1 M).

Sample Pretreatment

The quantitative recovery of TXA and GSH from the emulsion matrices (o/w and w/o) was performed through the following procedure: A specified volume of each sample (0.5 mL) was transferred into a glass container, followed by the addition of 2 mL of H_2O and 0.5 g of NaCl to induce emulsion breaking. The mixture was subjected to sonication and stirring for 10 min. Subsequently, two purification cycles were carried out using liquid-liquid extraction with 5 mL of diethyl ether to remove the oily phase. Each extraction step involved vortexing for 5 min and centrifugation for 10 min at 5000 rpm. After discarding the organic layer, 250 µL of the remaining aqueous phase were collected, diluted with water to a final volume of 50 mL, and analyzed using the proposed HPLC-FLD method.

Ex Vivo Permeability Study

Ex vivo permeation studies were performed using vertical Franz diffusion cells to evaluate and compare the transdermal permeation of glutathione (GSH) and tranexamic acid (TXA) incorporated into two different formulations, namely water-in-oil (w/o) and oil-in-water (o/w) emulsions, across porcine skin. Skin samples were mounted between the donor and receptor compartments with the stratum corneum oriented toward the donor side. The effective diffusion area was 0.2 cm², and the receptor compartment volume was approximately 5 mL. The receptor chamber was filled with phosphate-buffered saline (PBS, pH 7.4) and maintained at 37 °C under continuous stirring at 100 rpm to ensure sink conditions. The donor compartment was loaded with 0.1 mL of the respective emulsion corresponding to 100 µg of each API. The donor compartments were sealed with Parafilm

throughout the 24 h experiment to prevent drying of the skin tissue and loss of the applied formulation. At predetermined time intervals (1, 2, 4, 6, 8 and 24 h), 300 µL aliquots were withdrawn from the receptor compartment and immediately replaced with an equal volume of pre-warmed PBS. Samples were centrifuged prior to analysis, and the concentrations of GSH and TXA were quantified using HPLC. All experiments were conducted in triplicate ($n=3$).

The cumulative amount of drug permeated per unit area (Q_t, µg/cm²) was calculated according to Equation:

$$Q_t = \frac{1}{A}\left[V_r C_t + \sum_{n=1}^{t-1} V_n C_n\right]$$

where V_r is the receptor volume, C_t is the drug concentration at time t, V_n is the volume of the withdrawn sample, C_n is the drug concentration at the nth sampling point, and A is the effective diffusion area.

The steady-state flux (J_{flux}, µg cm^{-2} h^{-1}) was determined from the linear portion of the cumulative permeation profile using linear regression across all experimental time points. The apparent permeability coefficient (P_{app}, cm·h^{-1}) was calculated according to Equation:

$$P_{\text{app}} = \frac{J_{flux}}{C_0}$$

where, C_0 is the initial drug concentration in the donor compartment. P_{app} values were used to quantitatively compare the permeability of GSH and TXA between w/o and o/w emulsions.

At the end of the permeation experiment, the residual formulation remaining in the donor compartment was carefully collected using a pipette. The skin surface was subsequently rinsed with 0.1 mL of PBS and gently dabbed with absorbent tissue paper to remove any remaining surface residue prior to tape stripping.

Tape-stripping studies

Tape-stripping was employed to selectively remove the stratum corneum (SC) following completion of the permeation experiments, enabling assessment of drug distribution within the outermost skin layer. Although the effective diffusion area of the Franz cells was 0.2 cm² (diameter 5 mm), the skin samples mounted between the donor and receptor compartments were considerably larger than this diffusion area; following disassembly of the Franz cells, tape stripping was therefore performed on the whole excised skin piece, with adhesive tapes cut to match the dimensions of the skin sample, ensuring consistent and reproducible sampling across all replicates, as previously reported [27, 28]. All tape-stripping steps were performed by the same operator under standardized pressure to minimize variability. A standardized protocol was applied to all samples to minimize variability.

After the permeation studies, twenty adhesive tape strips (Scotch® Magic™ Tape 810, 3 M, France) were cut to match the diameter of the skin area and applied sequentially. Each strip was pressed firmly onto the skin using a roller passed five times in alternating directions to ensure uniform adhesion. The strips were then rapidly removed and placed into pre-labeled vials containing water, grouped as follows: vial 1 (strip 1), vial 2 (strips 2–3), vial 3 (strips 4–5), vial 4 (strips 6–8), vial 5 (strips 9–12), vial 6 (strips 13–16), and vial 7 (strips 17–20) [29]. The collected strips were sonicated for 1 h and centrifuged at 10,000 rpm for 10 min to extract the retained APIs. The resulting supernatants were analyzed by HPLC. Extraction efficiency was assessed using drug-spiked blank tapes processed under identical conditions.

Following SC removal, the remaining skin tissue was cut into small pieces and transferred to 15 mL centrifuge tubes. Extraction was initiated by adding 2 mL of methanol, followed by sonication for 15 min and vortexing for 15 s. Subsequently, 8 mL of water were added, and the mixture underwent two successive cycles of vortexing (15 s) and sonication (15 min). The samples were then placed under magnetic stirring for 30 min, followed by two additional cycles of vortexing (15 s) and sonication (15 min). A 1 mL aliquot of the final extract was collected, derivatized according to the proposed method, and analyzed by HPLC. All experiments were performed in triplicate ($n=3$).

Results and Discussion

Nano-emulsion Formulation

For the preparation of both emulsion types (o/w and w/o), a surfactant system and the corresponding aqueous and oil phases were required (Table 1). Span 80 and Tween 80 were used as emulsifiers at a ratio of 95:5. The aqueous phase consisted of water and Tween 80, while the oil phase contained Span 80 and one of the following oils: fractionated coconut oil with caprylic, capric, and lauric triglycerides; *Prunus armenica* kernel oil; or *Prunus amygdalus* dulcis oil. The concentration of the two active ingredients, TXA and GSH, was set at 1 mg/mL, following the composition of a commercial product (Eye Cream, Mary & May; Tranexamic Acid+Glutathione). Since both active compounds are water-soluble, they were first incorporated into the aqueous phase under stirring for 5 min, followed by sonication for an additional 5 min.

Subsequently, two mixing methods for combining the aqueous and oil phases of the emulsions were evaluated:

Table 1 Composition of w/o and o/w emulsions

Type	Oil (mL/gr)	Water (mL/gr)	Tween 80 (mL/gr)	Span 80 (mL/gr)	TXA (mg)	GSH (mg)
o/w	40/38	50/50	0.5/0.54	9.5/9.4	100	100
w/o	10/9.5	80/80	0.5/0.54	9.5/9.4	100	100

(a) ultrasound homogenization using a Sonics Vibra-Cell device (15 min, amplitude 65%, pulse 30/15), and (b) high-shear mixing using an Ultra-Turrax homogenizer (15 min) at 9,000 rpm [30]. In both cases, mixing was performed at low temperature using an ice bath. To determine the optimal mixing technique as well as the most suitable oil, all possible emulsion combinations were prepared and evaluated based on their zeta potential values and droplet size over a period of two weeks (Table 2).

According to the results, in all cases the absolute value of the zeta potential exceeded 30 mV, indicating good emulsion stability [31]. The droplet size of the emulsions prepared using the Ultra-Turrax was consistently larger than that of those produced with the Sonics Vibra-Cell system; therefore, the latter technique was selected as the more suitable method.

For the o/w emulsions, almond oil resulted in droplet sizes approximately three times larger compared to coconut and apricot oils. Conversely, in the w/o emulsions, apricot oil produced dispersed-phase droplets nearly twice as large as those obtained with the other two oils. Considering both droplet size and dispersion stability, fractionated coconut oil was identified as the optimal choice.

At this point, it should be noted that the present study was not aimed at formulation development or long-term formulation stability assessment. The formulations were used as vehicles for the comparative evaluation of GSH and TXA skin delivery under ex vivo conditions. Moreover, the emulsions were classified as nano-emulsions based on a broader droplet size criterion of 500 nm. This choice reflects the lack of a single universal cutoff in the literature, where nano-emulsions are variously defined as systems with droplet sizes of < 100 nm, 20–200 nm, or, in broader formulations, up to 500 nm or even 1000 nm [32–35]. Accordingly, the term nano-emulsion is used here in an inclusive sense, while recognizing that narrower definitions are also applied.

Optimization of the derivatization reaction

The OPA (o-phthalaldehyde) derivatization reaction is a chemical process in which OPA reacts, at ambient temperature (25 °C), with primary amines and thiols to form highly fluorescent isoindole derivatives. Although the reaction is almost instantaneous and the product could theoretically be detected fluorometrically, its yield and stability depend primarily on the type, concentration, and pH of the nucleophilic reagent used. In the present study, a series of parameters that could influence the intensity and stability of the analytical signal (Area Under the Curve, AUC) of the two analytes, were systematically investigated.

First, the appropriate solvent for preparing the stock solutions of TXA and GSH had to be selected to ensure complete solubility, promote the derivatization reaction, and provide satisfactory chromatographic peak shapes. Since both compounds are highly soluble in water and, to a lesser extent, in methanol, the effect of each solvent was examined separately. Both solvents produced comparable AUC values and similar reaction kinetics (derivative stability under specific conditions) during the first 5 h. A slight difference was observed at 24 h, where methanol yielded marginally higher peak areas. However, water was ultimately selected as the most suitable solvent, as it provided superior chromatographic peak shapes for glutathione.

In parallel, the efficiency and stability of the derivatives formed from the reaction of OPA with four different nucleophilic reagents N-acetyl-L-cysteine (NAC), 3-mercaptopropionic acid (3-MPA), 2-mercaptoethanol (2-ME), and tiopronin (Thiola), were evaluated over a period of 0–5 h

Table 2 Zeta potential and droplet size dispersed-phase particles using different oils and preparation methods

Emulsion type	Sonics Vibra Cell: Zeta Potential (mV) /Particle Size (nm)		
	Prunus armenica kernel oil	*Prunus amygdalus* Dulcia oil	Fractionated Coconut oil
o/w	-51/260	-32/1000	-44/260
w/o	-44/700	-39/400	-31/500
Ultra turrax: Zeta Potential (mV) /Particle Size (nm)			
	Prunus armenica kernel oil	*Prunus amygdalus* Dulcia oil	Fractionated Coconut oil
o/w	-53/4400	-60/1300	-60/1300
w/o	-51/300	-49/2300	-33/400

(Fig. 1). The results showed an increase in AUC values for GSH over time, whereas TXA exhibited a gradual decrease, regardless of the nucleophile used.

Among the four nucleophiles, NAC emerged as the optimal choice, providing the highest AUC values for both analytes. Examination of the reaction kinetics revealed that maximum yield occurred at 2 h, during which GSH increased by 3.22-fold, while TXA showed minimal degradation (96% stability). Therefore, NAC was selected as the most suitable nucleophilic reagent, combining high derivatization efficiency with environmental friendliness and non-toxic, therapeutically relevant properties (Fig. 2).

To further optimize the derivatization procedure, different concentration levels of OPA and NAC (0.01 M, 0.02 M, and 0.04 M) were also evaluated. The use of OPA at the highest concentration (0.04 M) resulted in a decrease in AUC values by 15% for GSH and 5% for TXA, whereas the lowest concentration (0.01 M) was insufficient to complete the reaction in certain samples. Therefore, the intermediate concentration of 0.02 M was selected as the most appropriate.

Similarly, for NAC, GSH exhibited the lowest AUC values at the highest reagent concentration (0.04 M), showing a reduction of approximately 17%, while TXA produced 33% lower AUC values at the lowest NAC concentration (0.01 M). Consequently, the concentration of 0.02 M was also identified as the optimal level for the nucleophilic reagent NAC.

Regarding the selection of the appropriate buffer, a borate solution was preferred due to its high buffering capacity [36]. In this context, one of the most critical parameters to be investigated was the pH, within the range of 8.5–11.5. According to the results, pH 10.5 was chosen as the optimum value, as increasing the pH led to a slight but consistent increase in AUC values, with the exception of the transition from pH 11 to 11.5, where the peak area of TXA was reduced by half.

The most critical aspect of this derivatization reaction, which proceeds under ambient conditions (25 °C) and is completed within 120 min, was to identify the optimal conditions that would ensure maximum stability of the two reaction products for as long as possible. To achieve this, the effects of temperature and pH (8–13) were systematically examined, either individually or in combination with storage under low temperature (4 °C), with the aim of stopping the reaction and stabilizing the derivatives.

In all experiments, samples were first kept at room temperature (25 °C) for 120 min to allow the reaction to reach

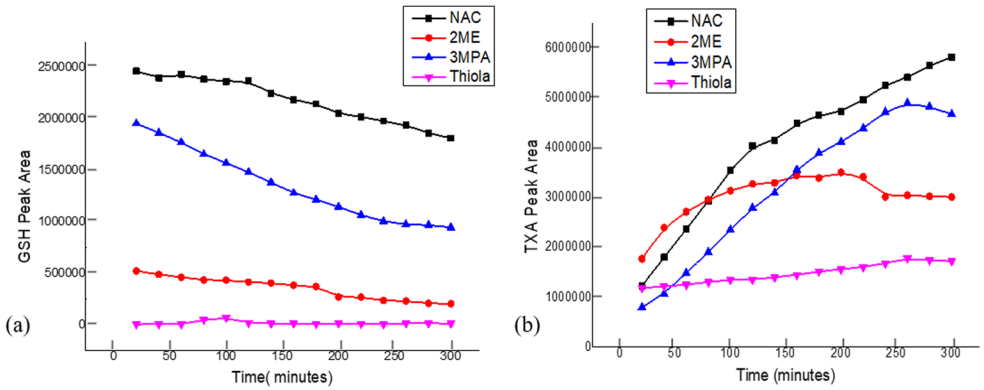


Fig. 1 Stability and efficiency study using different derivatization reagents for (**a**) GSH and (**b**) TXA

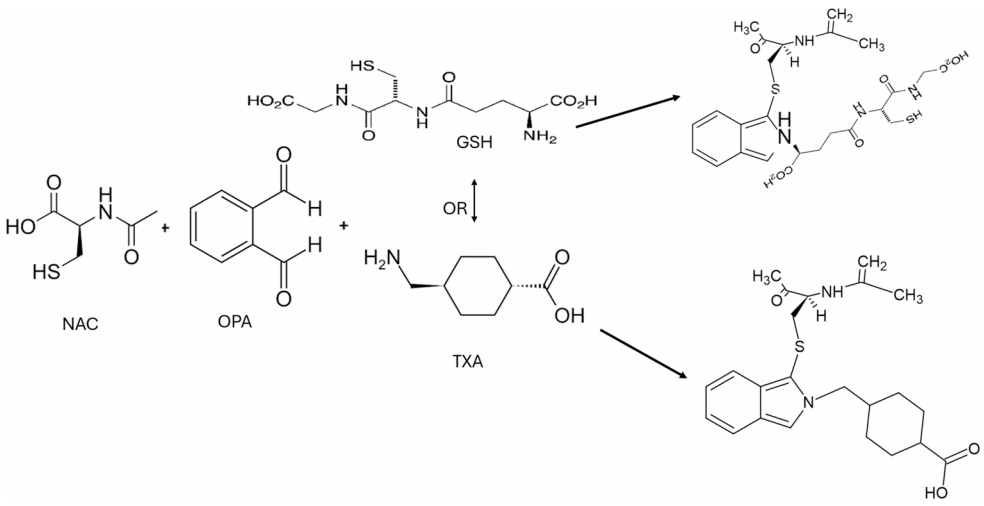


Fig. 2 Derivatization reaction of TXA or GSH with OPA in presence of NAC

completion. Subsequently, for the next 60 min, four different samples were placed at 60 °C, 25 °C, 4 °C, and 18 °C, respectively, and the AUC values of both analytes were measured at 20, 40, 60, and 80 min. The results showed that at higher (60 °C) and at room temperature (25 °C), the GSH derivative exhibited a continuous increase, whereas the TXA derivative showed a pronounced decrease. In contrast, when the temperature was maintained at 4–18 °C, both derivatives remained relatively stable for at least 3 h. However, at -18 °C the chromatographic peaks of GSH displayed poor peak shape, resulting in increased variability (RSD% > 4%). Therefore, maintaining the samples on a cooling rack (4 °C) throughout the analysis was deemed essential.

To investigate the effect of pH, after the initial reaction period, an appropriate volume of 0.1 N HCl was added to adjust the final pH to 8.2 (instead of the theoretically ideal 8.5 reported in the literature). Under these conditions, only the GSH derivative remained stable, while the TXA derivative degraded. Increasing the pH above 8.5 prevented TXA degradation and yielded stable AUC values but resulted in a significant increase in the GSH signal. For this reason, pH values up to 13 were also tested (using 0.1 N NaOH). At high pH, both GSH and TXA derivatives remained stable for at least 1 h, and when samples were simultaneously stored at 4 °C, stability extended up to 15 h.

Summarizing the procedure, the following steps were established: 450 µL of the aqueous sample solution were mixed with 50 µL OPA (0.02 M), 50 µL NAC (0.02 M), and 50 µL borate buffer (pH 10; 700 mM) in a clean glass HPLC vial. The mixture was vortexed for 15 s and allowed to react for 120 min at room temperature. After the addition of 125 µL NaOH 1 M, the sample was injected into the HPLC system (cooled rack at 4 °C) for analysis. Blank samples were prepared in parallel by replacing tranexamic acid and glutathione with deionized water.

HPLC Method Optimization

During the development of the chromatographic method for the determination of the two OPA complexes with GSH and TXA, the main challenge was to separate the glutathione peak from the solvent front, while ensuring that the tranexamic acid complex eluted within a reasonably short time. Several columns were tested, including C18 columns of different lengths (Supelco Discovery® HS 250–150 mm × 4.6 mm, 5 µm), a phenyl (Phenyl-ACE® 150 mm × 4.6 mm, 5 µm), and a CN column (Waters Spherisorb® 250 mm × 4.6 mm, 5 µm). The results showed that the only effective way to shift the glutathione peak away from the solvent front was by using a C18 column with 10% methanol as the organic modifier. However, these conditions significantly delayed the elution of the OPA-TXA complex, indicating the need for further optimization.

Considering that the two complexes share a large portion of their structural features, the effect of their ionization state was investigated. Both complexes contain a carboxylic group, while GSH additionally possesses two secondary amines and a thiol group. The use of an acidic mobile phase (H_2O-formic acid 0.2%) further accelerated the elution of GSH, which was undesirable. Therefore, a phosphate buffer (NaH_2PO_4) was employed to explore higher pH values (4.5–7.4).

As expected, pH 4.5 failed to separate glutathione from the solvent front and produced a broad peak for TXA. At pH 5.8, the separation between the two analytes deteriorated, with GSH eluting later and TXA earlier. pH values above 6.7 shifted GSH further away from the solvent front, but as the pH approached 7.4, the AUC of GSH decreased significantly. Moreover, the more alkaline environment did not accelerate the elution of TXA. For these reasons, pH 6.7 was selected as the optimal value.

Finally, the effect of the NaH_2PO_4 buffer concentration (5–50 mM) on the peak symmetry tailing factor) and retention times of the two complexes was investigated. The study showed that concentrations between 5 and 20 mM caused a slight increase in the retention times of both complexes, whereas concentrations from 25 to 40 mM produced no further changes. For these reasons, and to avoid excessive salt loading of the system, a concentration of 20 mM was selected.

The mobile phase flow rate was set at 1 mL/min, as it allowed the fastest possible analysis without overloading the system (pressure < 2000 psi throughout the run). Using a column oven at 30 °C slightly improved the peak shape of glutathione, while higher temperatures did not further enhance the chromatographic performance.

The different elution strength requirements of the two analytes led to the development of a gradient elution program. Initially, a low percentage of organic solvent (10% methanol for the first 5.5 min) was necessary to delay the elution of the glutathione complex. Subsequently, increasing the methanol content to 50% for 7 min accelerated the elution of the TXA complex. Approximately 7 min were required for the column to re-equilibrate to the initial conditions after the elution of the last peak.

HPLC Method Validation

The validation of the chromatographic method was carried out in accordance with the International Conference on Harmonization guidelines for the validation of analytical procedures (ICH Q2) [37]. The following parameters were

evaluated to confirm the method's reliability and suitability for quantitative analysis:

System Suitability

Since verifying the system's suitability is a fundamental requirement prior to initiating routine analysis, the appropriate test was performed. The evaluation included the assessment of retention time, peak area repeatability, theoretical plates, tailing factors, and the resolution between the two analytes (Table 3). All parameters met the predefined acceptance criteria, confirming that the chromatographic system was functioning properly and was suitable for subsequent analytical runs.

Specificity

To assess the specificity of the method, several blank solutions were prepared and analyzed to ensure that no chromatographic interferences occurred at the retention times of the two complexes. Specifically, the following blanks were prepared: (a) solutions containing only the derivatization reagents (OPA, NAC, and borate buffer) and water, (b) the emulsion base after undergoing extraction and derivatization, and (c) PBS in which porcine skin had been incubated and subsequently derivatized. Carryover was evaluated by three successive injections of the highest calibration standard followed by a blank sample injection. No detectable peaks were observed in the blank chromatogram, confirming the absence of carryover.

The chromatograms of all blank samples showed no interfering peaks at the retention times of the analytes, demonstrating that the method exhibits excellent specificity. (Fig. 3).

Linearity, limit of detection (LOD) and limit of quantitation (LOQ)

Linearity was evaluated by constructing calibration curves for each analyte using at least six concentration levels. The linear relationship between analyte concentration and peak area (AUC) was assessed by least-squares regression analysis, and the correlation coefficients (R^2) exceeded 0.999 for both GSH and TXA (Table 4). To further confirm the quality of the fit, residual plots were examined to ensure homoscedasticity across the calibration range. The formulas LOD = 3 S/N and LOQ = 10 S/N were used for the calculation of the limits of detection (LOD) and quantitation (LOQ), respectively, where S = signal and N = noise [38].

Precision

To evaluate precision, both repeatability and intermediate precision were assessed. For repeatability, a standard solution at the lower concentration level of each analyte was analyzed in six replicates within the same day. The %RSD

Table 3 System suitability parameters

Analytes	$\%RSD_{AUC}$ ($n=6$)	Tr* (mins)	Tf*	Rs*	*N**	HETP* (µm)
GSH	0.95	5.87	1.36	first eluting peak	2,524.5	349,830
TXA	1.59	12.19	1.58	18.2	46,578.2	1,893,806

**Tr* Retention Time, *Tf* tailing factor, *Rs* resolution, *N* number of theoretical plates, *HETP* height equivalent of a theoretical plate

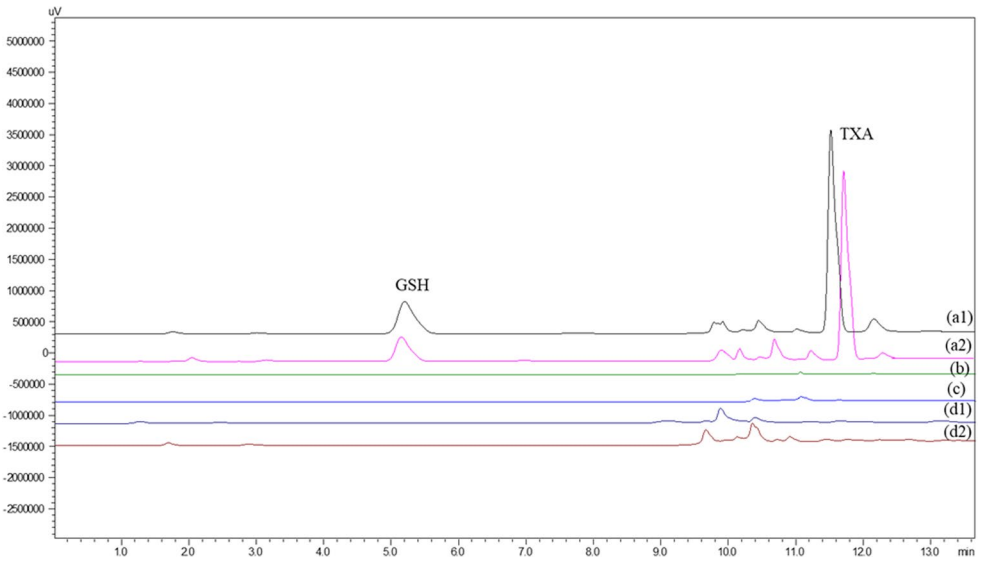


Fig. 3 Chromatograms of: (**a**) standard extracted nano-emulsion (1) w/o and (2) o/w (**b**), blank water (**c**) blank PBS at Tissue and (**d**) blank emulsion after extraction (1) w/o and (2) o/w

Table 4 Linear regression analysis data

Analytes	Concentration Range (ng/mL)	Equation	(R^2)	LOD (ng/mL)	LOQ (ng/mL)
GSH	7.5–375	y= (14604.7 ± 61) x − 32,691 ± 11,611	0.999	2.6	8.0
TXA	7.5–375	y = (29847.7 ± 184) x − 14,979 ± 35,179	0.999	3.9	11.8

values obtained were < 1.6%, indicating excellent intra-day precision. For intermediate precision, samples at three concentration levels (low, medium, and high) were prepared and analyzed over three consecutive days, with three replicates per level. In all cases, the %RSD values were < 2.25%, confirming the robustness of the method under normal operating conditions (Table 5).

Accuracy

To evaluate the accuracy of the analytical method, five freshly prepared standard solutions of the two analytes at known concentrations were analyzed after derivatization. The measured concentrations were calculated using the corresponding calibration curve (Table 6). The accuracy of the sample clean-up procedure (emulsion matrix) and the recovery of the two analytes from the substrate is assessed separately in Sect. 3.5, where the extraction method is described in detail.

Robustness

The robustness of the method was evaluated by monitoring the AUC and tailing factor (Tf) of the analytes under small deliberate variations of key chromatographic parameters, including column temperature, mobile phase flow rate, mobile phase composition, pH, and detection wavelength. The method was considered robust, as all %RSD values were below 2% under every tested condition (Table 7).

Table 6 Accuracy results for GSH and TXA based on five independently prepared standard solutions

Concentration (ng/mL)	GSH		TXA	
	Found	%Recovery	Found	%Recovery
7.5	7.31	97.5	7.80	104.0
37.4	36.89	98.4	38.35	102.2
75.0	72.50	96.7	76.96	102.6
187.5	194.25	103.6	182.09	97.1
375.0	372.00	99.2	377.25	100.6

Table 7 Robustness test of the proposed analytical method

Parameters	GSH		TXA	
	AUC (% RSD)	Tf	AUC (%RSD)	Tf
Flow rate (0.95, 1.0, 1.05 mL/min)	0.83	0.82	1.79	0.92
Temperature (28, 30, 32 ◦C)	0.98	0.71	0.86	0.78
Mobile phase B at gradient (9:49, 10:50, 11:51 v/v)	1.34	0.57	0.97	0.98
λ_{exc} (420,425,430 nm)	1.81	0.53	1.74	0.21
pH mobile phase (6.5, 6.7, 6.9)	1.2	0.17	1.31	0.17

SALLE Extraction of GSH and TXA from Emulsions

Both GSH and TXA are highly water-soluble compounds (20 mg/mL and 31 mg/mL, respectively) [39, 40]. Due to this characteristic, water was selected as the most suitable solvent for their extraction from the emulsion. However, in this specific emulsion system, TXA and GSH cannot be efficiently extracted using water and conventional liquid-liquid extraction technique, as the resulting solution lacks sufficient clarity. Therefore, the recovery of the analytes was based on a salt-assisted liquid-liquid extraction (SALLE) technique, which provided improved phase separation and clarity [41].

Specifically, 0.5 mL of the formulation was diluted to a final volume of 2.5 mL with water. Subsequently, 0.5 g of NaCl was added, and the mixture was subjected to magnetic stirring followed by ultrasound treatment for a total of

Table 5 Repeatability and intermediate precision analysis

APIs	Repeatability		Intermediate Precision				
	Concentration (ng/mL)	RSD%	Concentration (ng/mL)	1st day	2nd day	3rd day	RSD%
GSH	7.5 (n=6)	0.95	7.5 (n=3)	0.95	0.91	1.48	2.22
	75 (n=3)	1.38	75 (n=3)	1.38	1.37	1.57	1.84
	375 (n=3)	0.51	375 (n=3)	0.51	1.91	1.92	1.72
TXA	7.5 (n=6)	1.59	7.5 (n=3)	1.59	1.09	1.59	1.79
	75 (n=3)	1.35	75 (n=3)	1.35	0.62	1.15	1.72
	375 (n=3)	0.99	375 (n=3)	0.99	1.31	1.29	1.68

10 min. Sample cleanup was achieved by adding 2.5 mL of an organic solvent. The solvents tested were dichloromethane, ethyl acetate, hexane, chloroform, and diethyl ether. Hexane was excluded because it failed to form a clear aqueous layer even after 72 h or after centrifugation, making it unsuitable for producing clean samples for analysis. The remaining four solvents produced a clear aqueous phase after centrifugation (10 min, 5000 rpm), which was appropriate for further analysis.

A 100-µL aliquot of the aqueous layer was diluted to a final volume of 20 mL with water. After derivatization according to the proposed procedure, the samples were analyzed by HPLC-FLD. The selection of the optimal organic solvent was based on the recovery values obtained from the formulation. As shown in Table 8, diethyl ether provided the highest recoveries, ranging from 97.7% to 103.8% (RSD% < 3, $n = 5$).

Analytical Assessment

The Modified Green Analytical Procedure Index (MoGAPI) was used to evaluate the environmental impact and safety of the analytical method by considering sample pretreatment, reagent and solvent consumption, as well as instrumentation requirements. The obtained MoGAPI score of 60 indicates that the proposed method can be considered environmentally acceptable, with reduced chemical usage and waste generation (Fig. 4a) [42]. The Blue Applicability Grade Index (BAGI) assessed the practical applicability of the method based on ten criteria related to White Analytical Chemistry, including feasibility and operational performance. A BAGI score of 65 demonstrated that the protocol possesses strong applicability potential for routine analytical use (Fig. 4b) [43]. In parallel, the Red Analytical Performance Index (RAPI) was employed to evaluate the analytical performance of the method with respect to selectivity, sensitivity, accuracy, precision, reproducibility, analysis time, and sample preparation complexity. The resulting RAPI score of 92.5 confirmed the high overall performance and reliability of the proposed procedure (Fig. 4c) [44]. Finally, the Click Analytical Chemistry Index (CACI) assessed the method according to the principles of click chemistry, emphasizing simplicity, efficiency, reliability, portability, and automation. The achieved CACI score of 73% was considered satisfactory, indicating that the method combines acceptable analytical performance with practical simplicity and applicability (Fig. 4d) [45].

Table 8 Recovery values of GSH and TXA obtained using different organic solvents in the SALLE procedure

Organic Solvent	GSH		TXA	
	%Recovery (o/w)	%Recovery (w/o)	%Recovery (o/w)	%Recovery (w/o)
Dichloromethane	70.04	72.34	103.13	134.35
Diethyl Ether	100.57	103.81	97.69	97.74
Ethyl Acetate	85.71	105.77	86.10	98.64
Chloroform	15.12	30.52	37.6	135.72

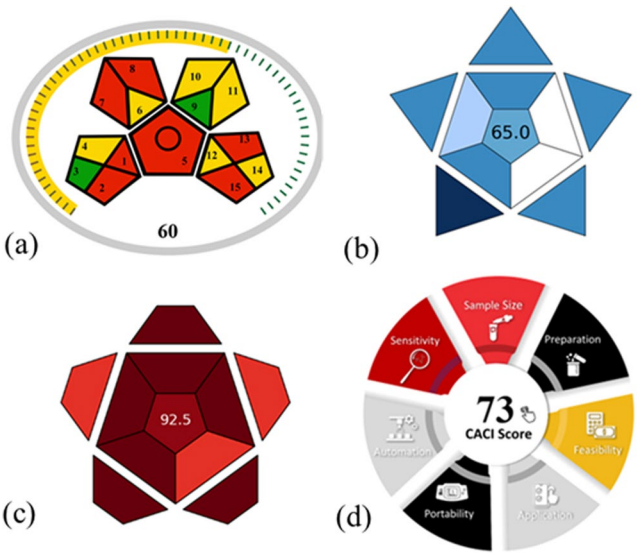


Fig. 4 Analytical method assessment with (**a**) MoGAPI, (**b**) BAGI, (**c**) RAPI and (**d**) CACI

Quantitative Recovery Assessment from Tissue and Tape Stripping Samples

To ensure accurate quantitative recovery of the active compounds from both Franz diffusion cell tissue and tape-stripping samples, a systematic evaluation of the two substrates was performed. The procedure proved particularly demanding, mainly in achieving precise and reproducible spiking of the substrates.

In order to spike the tape strips, three solutions (diluent water) of different concentrations (10,100 and 350 µg/mL) were prepared, and 10 µL of each were applied onto the tape surface and allowed to dry completely ($n = 3$ per level).

Verification of the spiked amounts within the tissue was likewise conducted at three concentration levels, with three replicates each. The tissue was placed in an airtight vial containing 10mL of the active compounds solutions (100, 200 and 350 ng/mL) and kept under continuous stirring at room temperature (25 °C) for 24 h to allow diffusion of the substances into the tissue. After 24 h, both the liquid phase (after filtration) and the extracted tissue sample were analyzed and compared with the initial concentrations. Mass balance calculations enabled determination of the exact amount absorbed by the tissue and confirmed quantitative recovery.

The extraction procedure began with the addition of 2 mL MeOH to the spiked tissue or tape, followed by 15 min of sonication and 15 s of vortex mixing. Water was then added to a final volume of 10 mL, and the mixture underwent two successive cycles of vortexing (15 s) and sonication (15 min). This was followed by 30 min of magnetic stirring and a repetition of the two vortex-sonication cycles. A 1 mL aliquot of the final extract was derivatized and analyzed using the proposed HPLC-FLD method.

The results demonstrated recovery values greater than 95% across all concentration levels, with RSD < 4% for both active compounds, confirming that the extraction procedure is reliable for the quantitative evaluation of samples obtained from Franz diffusion and tape-stripping experiments.

Stability studies

To investigate the stability of the active compounds either within Franz diffusion cells or in formulation form under storage conditions (at 37 °C), the following procedure was carried out. Initially, solutions of GSH and TXA (1 mg/mL) were prepared in PBS as well as in o/w and w/o emulsions. These samples were incubated in a water bath at 37 °C. Aliquots were collected at 1, 2, 4, 6, 8 and 24 h, and after appropriate sample preparation, both analytes were quantified. According to the results (Fig. 5), GSH and TXA exhibited thermal instability in PBS and in both emulsion bases. PBS provided the highest stability, whereas the o/w emulsion showed the lowest. Between the two analytes, GSH demonstrated markedly greater instability, decreasing to approximately 50% of its initial concentration after 24 h, TXA showed losses of approximately 8% over the same period, reflecting its considerably greater thermal stability compared to GSH. During the first two hours, all samples exhibited minor losses, averaging around 10%. These findings provide an indication of the extent to which the two compounds may degrade when entering systemic circulation, upon exposure to sunlight, or during storage at elevated temperatures.

Finally, a stability study of the two formulations (o/w and w/o), for a short period of time, was conducted over a period of one month at 4 °C. Under these storage conditions, the emulsion bases maintained stable zeta potential, droplet size, and active compound concentrations. Therefore, the formulations appear suitable as potential vehicles for the development of a pharmaceutical or cosmetic product.

Ex vivo permeation studies

The cumulative permeation profiles of GSH and TXA from w/o and o/w emulsions across porcine skin are shown in Fig. 6a and b, respectively.

In the present study, tranexamic acid demonstrated marked formulation-dependent differences in skin permeability. Owing to its relatively low molecular weight (~ 157 Da) and hydrophilic nature, tranexamic acid can diffuse efficiently once it partitions into the stratum corneum, making its transdermal transport highly sensitive to the composition and architecture of the applied emulsion system [46]. Due to its pronounced hydrophilicity, tranexamic acid exhibits limited affinity for the lipid-rich domains of the skin, which can restrict its interaction with the stratum corneum lipid matrix [47].

Application of TXA in the w/o emulsion (Fig. 6a) resulted in significantly higher cumulative permeation after 24 h (190.51 ± 7.24 µg/cm²) compared to the o/w emulsion (105.87 ± 5.72 µg/cm², Fig. 6b), corresponding to an approximately 1.8-fold enhancement in permeation for the w/o formulation. This difference was further reflected in the calculated permeability parameters, with a higher apparent permeability coefficient for TXA in the w/o emulsion (P_{app} = 17.62 ± 3.14 × 10^{-3} cm/h) compared to the o/w formulation

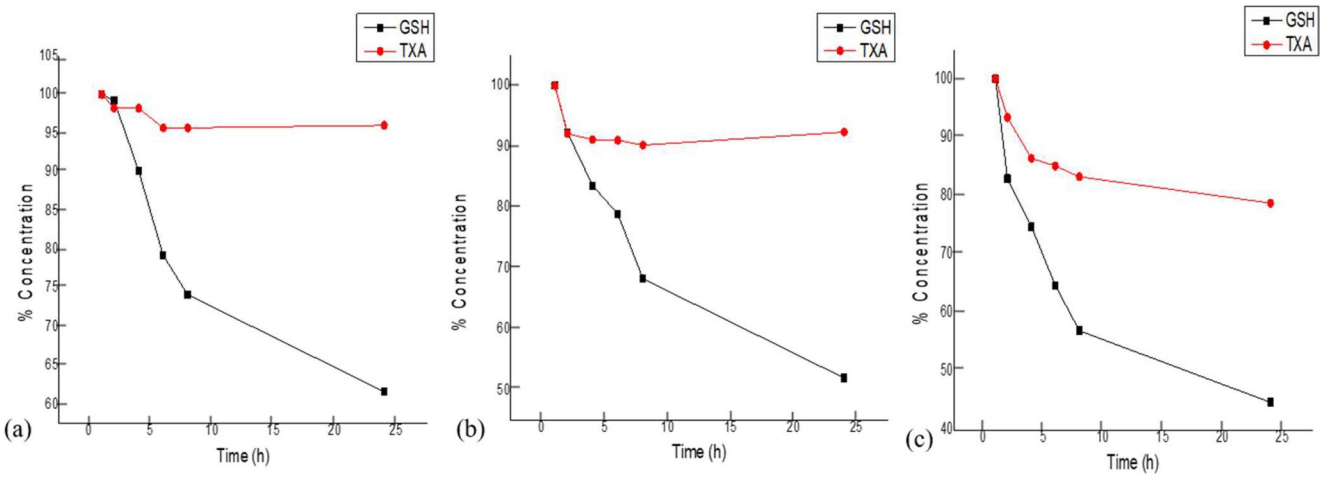


Fig. 5 Stability test at 37 °C in: (**a**) PBS, (**b**) w/o emulsion and (**c**) o/w emulsion base

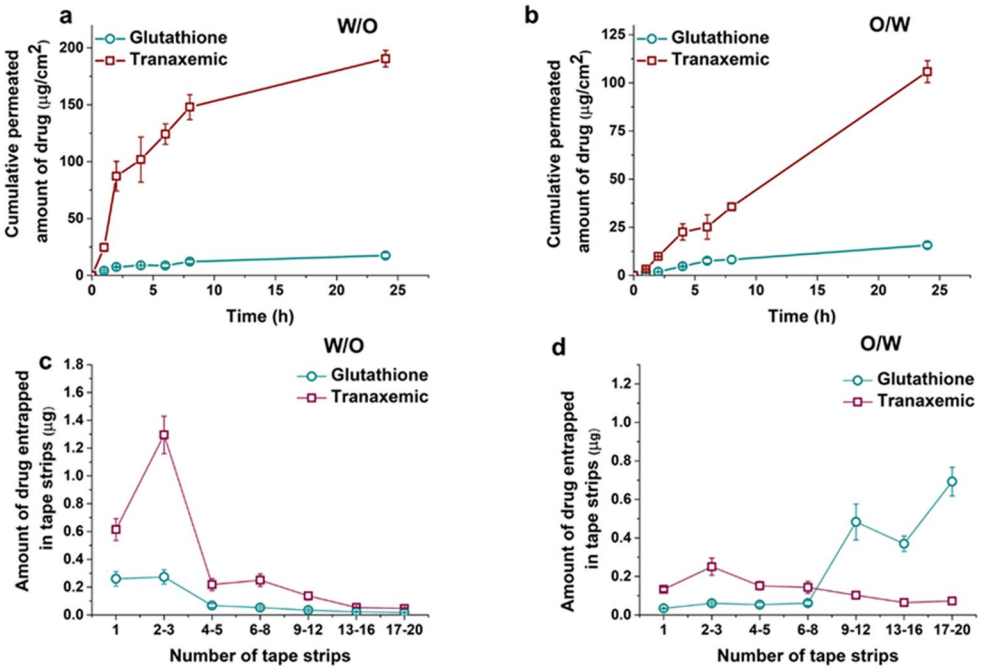


Fig. 6 Cumulative permeation data for (**a**) GSH and TXA (W/O) emulsion and (**b**) GSH and TXA (O/W) emulsion over 24 h across porcine skin (Mean ± S.D., $n = 3$). Tape stripping method employed to evaluate the amount of GSH and TXA detected in the stratum corneum after application of (**c**) W/O emulsion and (**d**) O/W emulsion on porcine skin after 24 h application

(P_{app} = 15.16 ± 2.19 × 10^{-3} cm/h), as summarized in Table 9. Analysis of the linear region of the TXA permeation profiles demonstrated a higher steady-state flux for the w/o formulation (J_{flux} = 5.66 ± 0.80 µg/cm²/h) compared to the o/w formulation (J_{flux} = 4.38 ± 0.24 µg/cm²/h). These findings indicate that the presence of an external oil phase enhances TXA transport across the skin.

This behavior is consistent with previous studies reporting that w/o emulsions, through prolonged contact of the oil phase with the SC, can promote drug partitioning into the intercellular lipid matrix and enhance diffusion by modulating lipid organization and fluidity [48, 49]. In contrast, o/w emulsions present an aqueous continuous phase at the skin surface, which facilitates rapid surface availability of TXA but limits sustained interaction with SC lipids [50]. As a result, penetration into deeper skin layers is reduced, leading to lower cumulative permeation and flux values over the same experimental period. The tapering of the TXA cumulative permeation profile observed at later time points in the w/o system is consistent with progressive depletion of the donor compartment, which reduces the thermodynamic driving force for permeation as the experiment proceeds.

In contrast to TXA, glutathione (GSH) exhibited very limited skin permeation from both emulsion types.

A decline in GSH levels is associated with various degenerative diseases and skin disorders [51].

However, GSH's instability and poor penetration through the skin's stratum corneum (SC) pose significant challenges for its topical application [52].

As shown in Fig. 6a and b, the cumulative amount of GSH permeated from the w/o emulsion (17.51 ± 2.26 µg/cm²) was comparable to that obtained from the o/w emulsion (15.69 ± 1.40 µg/cm²) after 24 h, indicating that GSH permeation was essentially independent of formulation type. Although a gradual increase in permeation was observed over time for both emulsions, the absolute amounts remained low, reflecting the strong barrier function of the stratum corneum (SC). This limited permeation is consistent with the highly polar, hydrogen-bond-rich structure of glutathione, which severely restricts its ability to diffuse through the

Table 9 Permeation parameters and skin deposition of glutathione and tranexamic acid following application of water-in-oil (W/O) and oil-in-water (O/W) emulsions across porcine skin

	J_{flux} (µg/cm²/h)*	P_{app} *10^{-3} (cm/h)*	Extracted amount of drug in tissue after tape stripping (µg)#
W/O- Glutathione	0.519 ± 0.122	3.385 ± 0.509	16.821 ± 0.927
W/O- Tranexamic	5.660 ± 0.799	17.622 ± 3.139	8.132 ± 0.996
O/W- Glutathione	0.594 ± 0.072	3.883 ± 0.453	8.452 ± 1.175
O/W- Tranexamic	4.376 ± 0.235	15.157 ± 2.192	10.042 ± 1.181

*Steady-state flux and apparent permeability coefficient were determined from Franz diffusion cell experiments

#The extracted amount of drug (µg) represents the total quantity recovered from the skin after completion of the tape-stripping procedure

Values are expressed as mean ± standard deviation ($n = 3$)

lipid matrix of the SC. As a result, the intrinsic barrier properties of the SC outweighed any formulation-related effects, and modifications in emulsion architecture were insufficient to produce a meaningful enhancement in flux.

This conclusion is further supported by the permeability coefficients (Table 9), where the P_{app} values for the o/w ($3.88 \pm 0.45 \times 10^{-3}$ cm/h) and w/o ($3.36 \pm 0.509 \times 10^{-3}$ cm/h) emulsions were nearly identical. In contrast, extraction analysis revealed pronounced formulation-dependent differences in skin retention. Specifically, the amount of GSH recovered from the skin after tape stripping was significantly higher for the w/o emulsion (16.82 ± 0.93 µg) compared to the o/w emulsion (8.45 ± 1.18 µg), corresponding to an approximately two-fold increase in tissue deposition. These findings demonstrate that, for GSH, formulation effects manifest primarily in terms of skin retention rather than transdermal permeation, with the external oil phase promoting greater deposition of the compound within the skin layers.

It should be acknowledged, however, that the degradation of GSH observed in the stability studies (Fig. 5) represents a potential confound that should be considered when interpreting permeability results. The introduction of stabilizing agents during the IVPT experiment, such as antioxidants, was deliberately avoided, as such additives may alter skin barrier integrity and compromise the physiological relevance of the permeation data. Consequently, the extent to which the observed low GSH permeation reflects true barrier limitation versus chemical instability within the donor and/or receptor compartments cannot be fully delineated from the present data. This represents an inherent limitation when studying chemically labile molecules such as GSH in ex vivo permeation models.

In summary, the contrasting permeability behavior between TXA and GSH in the two emulsions is not coincidental but is directly and predictably determined by their physicochemical properties, which define the extent to which each molecule can interact with the lipid matrix of the stratum corneum. TXA, with a molecular weight of approximately 157 Da and a log P of ~0.09, exhibits a borderline hydrophilic-lipophilic profile. This near-neutral lipophilicity provides sufficient amphiphilic character to enable partial partitioning into the intercellular lipid lamellae of the SC, particularly when the formulation establishes favorable thermodynamic conditions at the skin surface. The external oil phase of the w/o emulsion fulfils this role by maintaining prolonged oil-skin contact and reducing trans-epidermal water loss, thereby shifting the partitioning equilibrium in favor of TXA entry into the SC lipid domains and enhancing both SC uptake and transdermal flux. In contrast, the o/w system presents an aqueous continuous phase that ensures rapid surface availability of TXA but cannot sustain the drug-lipid interactions required for efficient SC partitioning, resulting in comparatively lower permeation.

GSH, on the other hand, is a tripeptide bearing two carboxylate groups, a primary amine, and a free thiol-functional groups that are effectively ionized at physiological pH. As a result, GSH possesses a strongly negative effective log P and an exceptionally high hydrogen-bonding capacity. These characteristics render GSH fundamentally incompatible with the lipid bilayer environment of the SC: the energetic cost of desolvating such a highly hydrophilic, multiply charged molecule prior to entry into the lipid phase is prohibitively high, and neither emulsion type provides a sufficient thermodynamic driving force to overcome this barrier. In addition, the mass balance results (40–46% total GSH recovery) are consistent with the ~50% degradation observed in the 24 h stability study at 37 °C, indicating that a substantial fraction of the applied dose was chemically lost during the IVPT experiment. Therefore, the negligible permeation of GSH cannot be attributed solely to its physicochemical incompatibility with the SC lipid matrix but also to its intrinsic instability under the experimental conditions, which reduces the amount of intact GSH available for diffusion. This dual limitation should be considered when interpreting the permeation profile of GSH in ex vivo systems.

Tape stripping studies

The epidermis and dermis are critical for the pharmacological efficacy of active ingredients in topical formulations. Effective topical drug delivery requires penetration of the skin barrier and delivery of the active compounds into the viable epidermis and dermis. Tape stripping is widely used for quantifying drug levels within the stratum corneum (SC) and for assessing formulation-dependent differences in SC uptake and distribution [52, 53].

Tape-stripping analysis (Fig. 6c and d) revealed clear formulation-dependent differences in the SC distribution of TXA. For both emulsion systems, the amount of TXA recovered from the tape strips decreased progressively with increasing SC depth, indicating predominant localization in the superficial layers. Furthermore, TXA delivered from the w/o emulsion was detected across a greater number of consecutive tape strips, extending into deeper SC layers.

The broader SC distribution observed for TXA following application of the w/o formulation aligns with the higher cumulative permeation and steady-state flux measured in the Franz diffusion studies, indicating enhanced uptake of the drug into the SC. This suggests that, in the present system, increased SC uptake was associated with more efficient subsequent diffusion across the skin. Previous studies have shown that emulsion architecture, particularly the presence of an oil-continuous phase, can significantly influence both drug release and skin transport, leading to formulation-dependent differences in cumulative permeation and flux. Such effects

are attributed to vehicle-dependent drug partitioning and interactions with the SC barrier, rather than to a universal relationship between penetration depth and flux [48].

Tape-stripping analysis revealed more distinct formulation-dependent effects for GSH at the level of SC distribution (Fig. 6c and d). A markedly different and scientifically noteworthy distribution pattern was observed for GSH applied in the o/w emulsion (Fig. 6d). Unlike the conventional decreasing concentration profile typically expected with increasing SC depth, GSH exhibited a progressive increase in the recovered amounts from the superficial toward the deepest sampled layers. Specifically, low quantities were detected in the outermost strips (strips 1–3: ~0.09 µg), which increased substantially at intermediate depths (strips 9–12: ~0.48 µg), reaching the highest levels in the deepest layers sampled (strips 17–20: ~0.70 µg).

This inverse depth-dependent distribution pattern suggests that the aqueous continuous phase of the o/w emulsion induced pronounced SC hydration, leading to corneocyte swelling and disruption of the ordered lamellar lipid architecture within the intercellular spaces. For a molecule as highly hydrophilic and rich in hydrogen-bonding groups as GSH, this structural loosening generates transient aqueous microchannels within the deeper SC layers, regions that naturally contain higher water content, along which GSH can redistribute by diffusion, progressively accumulating within the intracellular and intralamellar hydrophilic domains of the deeper corneocyte layers.

Importantly, this hydration-mediated redistribution did not translate into enhanced transdermal permeation, as the cumulative GSH flux remained negligible and comparable between the two formulations. This confirms that SC hydration facilitates inward redistribution of GSH within the SC but does not enable it to cross the lipid barrier and reach the receptor compartment. Hydration-induced corneocyte swelling and disruption of hydrogen bonding within keratin-rich intracellular domains have been reported to enhance the diffusion of polar solutes along intracellular and intralamellar pathways [49, 54]. Experimental tape-stripping studies have similarly shown that increased SC hydration can lead to greater recovery of hydrophilic compounds from deeper SC layers, while overall permeability across the skin barrier remains limited [29]. In contrast, glutathione applied in the w/o emulsion was predominantly retained within the superficial SC layers, with sharply decreasing amounts detected at greater depths. This behavior is consistent with reduced hydration-driven diffusion when an oil-continuous external phase is present, favoring superficial retention of highly hydrophilic compounds.

Taken together, the permeation and tape-stripping data presented in this study demonstrate that emulsion architecture exerts compound-specific effects on skin drug distribution, which cannot be generalized across molecules with different physicochemical profiles. For TXA, the w/o formulation provides a clear and consistent advantage at every analytical level: higher cumulative permeation, greater steady-state flux, enhanced SC uptake, and a broader depth distribution within the SC. These effects arise from the ability of the external oil phase to promote drug partitioning into the SC lipid matrix.

For GSH, formulation effects do not manifest in transdermal flux, which remains limited by the skin barrier and independent of formulation type, but rather in the spatial distribution of the molecule within the SC and in the total amount retained in the skin tissue. The w/o emulsion promotes superficial SC retention and higher overall tissue deposition due to its occlusive properties, whereas the o/w emulsion induces hydration-mediated redistribution of GSH into deeper SC layers without enabling transdermal passage.

These findings have direct implications for the rational design of topical formulations. For a borderline hydrophilic molecule such as TXA, selecting an oil-continuous vehicle can meaningfully enhance both SC uptake and transdermal delivery. In contrast, for a highly polar, ionised molecule such as GSH, conventional emulsion systems are insufficient to overcome the intrinsic barrier imposed by the SC lipids, and more advanced delivery strategies, such as lipid nanocarriers, vesicular systems, or physical enhancement techniques, would be required to achieve therapeutically relevant concentrations within the viable epidermis. Mass balance analysis further supported the validity of the permeation experiments. The total amount of GSH recovered across all compartments following w/o emulsion application averaged 45.70 ± 11.77 µg from 100 µg initially applied, consistent with the ~50% degradation observed in the stability studies conducted under identical experimental conditions. Similarly, TXA exhibited a total recovery of 92.83 ± 9.60 µg, in agreement with its higher thermal stability (~8% degradation under the same conditions). Comparable results were obtained following o/w emulsion application, where GSH and TXA recovery averaged 40.21 ± 3.39 µg and 77.18 ± 14.03 µg, respectively, values equally consistent with the degradation profiles established in the stability studies.

Conclusions

In the present work, a reliable, selective, and highly sensitive HPLC-FLD method was successfully developed and validated for the simultaneous determination of GSH and TXA in emulsions and ex vivo permeability studies, following derivatization with OPA/NAC. The method demonstrated excellent linearity ($R^2 > 0.999$), high accuracy and repeatability (RSD < 2.2%), and no interference from

blanks, while its robustness was confirmed through minor variations in chromatographic parameters. For sample preparation, the SALLE technique provided clean extracts and high recoveries, with diethyl ether yielding the best performance (97.7-103.8%). Overall, the method is fully suitable for routine analysis in complex matrices.

Stability studies in PBS and emulsions showed that both active compounds are thermally unstable at 37 °C, with PBS offering the highest relative stability and the o/w emulsion the lowest. GSH exhibited approximately 50% degradation within 24 h, whereas TXA demonstrated considerably greater thermal stability, with losses of only approximately 8% under the same conditions. In contrast, storage of the emulsions at 4 °C for one month demonstrated complete physicochemical and chemical stability, confirming their suitability as carriers.

Permeation and tape-stripping studies revealed that emulsion architecture affects TXA and GSH differently, depending on their physicochemical properties. TXA showed significantly enhanced uptake, deeper SC distribution, and higher transdermal transport with the w/o emulsion, due to its improved interaction with the SC lipid matrix. In contrast, GSH remained barrier-limited, with the w/o system increasing only superficial retention and the o/w system inducing hydration-mediated redistribution into deeper SC layers, without meaningful transdermal passage. These findings highlight that vehicle selection must be tailored to the physicochemical profile of each molecule to achieve optimal topical performance and therapeutic efficacy.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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